

From Monoculture to Market: The Economics of Plant Biodiversity, Systemic Risk, and Consumer Welfare

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Abstract

Modern industrial agriculture relies on monocultures to maximize economies of scale, yet this genetic homogenization introduces profound systemic risks to global food security. This paper investigates whether market competition naturally corrects this trend or exacerbates it, utilizing a two-stage structural model of endogenous cultivar choice and price competition. By estimating a Random Coefficients Logit model, I quantify the static welfare gains from cultivar variety and the dynamic insurance value of biodiversity against correlated supply shocks. Simulation results demonstrate that consumers derive significant utility from variety, with a transition to pure monoculture resulting in a 73% loss in static consumer surplus. Crucially, I find that market structure determines this vulnerability: a simulated competitive oligopoly sustains four distinct cultivars, whereas a monopoly restricts the market to just two, internalizing cannibalization. Furthermore, I identify a substantial market failure: while monoculture is statically efficient in normal states, it creates unpriced systemic tail risk. Under simulated shocks, the expected social welfare of a diverse biological portfolio is approximately 68% higher than that of a monoculture. These results suggest that profit-maximizing firms structurally under-provide biodiversity, providing a rigorous economic rationale for biodiversity subsidies and stringent antitrust scrutiny in agribusiness.

1 Introduction

The trajectory of modern industrial agriculture has been defined by a relentless pursuit of yield and uniformity. Driven by economies of scale in harvesting, packing, and global shipping, agricultural markets have converged on a narrow set of highly productive cultivars—a phenomenon known as monoculture. For example, the Cavendish banana accounts for over 99% of the global export market. While this homogenization has successfully lowered marginal costs and consumer prices in the short term, it has rendered the entire global supply chain critically vulnerable to specific pathogens, such as the Tropical Race 4 (TR4) fungus.

This paper asks whether free market competition naturally corrects this monoculture trend or structurally exacerbates it. Market competition exerts two opposing forces on biodiversity. The Differentiation Force encourages firms to escape fierce price competition by introducing unique, niche cultivars to capture specific consumer tastes, theoretically improving biodiversity. Conversely, the Efficiency Force drives intense cost-cutting; because supply chains are optimized for uniform products, firms chase scale economies by converging on a single high-yield cultivar, thereby destroying biodiversity. To evaluate these forces, this research asks: How does market structure affect the equilibrium provision of biodiversity? What is the static consumer surplus generated by cultivar variety? And critically, what is the social welfare gap created by the market’s failure to price the systemic tail risk of monoculture?

Answering these questions is of first-order importance for both economic policy and global food security. Standard economic analyses often view agricultural output as homogeneous, measuring efficiency solely through yield and price. However, biological uniformity creates a massive, unpriced negative externality: systemic tail risk. A novel climate shock or pathogen adapted to the dominant cultivar could trigger a catastrophic collapse of the food supply. Quantifying this dynamic insurance value of biodiversity translates ecological resilience into a rigorous economic metric, providing actionable parameters for antitrust regulators evaluating agribusiness mergers and policymakers designing seed-diversity subsidies.

To address this, I develop and estimate a two-stage structural Industrial Organization (IO) model using high-frequency retail scanner data mapped to biological cultivars. In Stage 1, firms endogenously choose which cultivars to supply, incurring fixed costs of adoption. In Stage 2, firms compete on price in a Bertrand-Nash equilibrium. Consumer demand is modeled via a Random Coefficients Logit (BLP) framework, allowing for heterogeneous preferences for crop attributes like sugar content and disease resistance. The parameters are estimated using Generalized Method of Moments (GMM), employing the characteristics of

competing cultivars as instrumental variables.

The simulated results reveal a profound economic tension and a significant market failure. Structural estimation successfully recovers underlying preferences, confirming a robust static consumer willingness-to-pay for variety. Specifically, the model shows that consumers are willing to pay a \$0.33 mean premium for a one-unit increase in quality (sugar content), and that removing variety leads to a 73% loss in consumer surplus. Addressing the first research question, I find that market structure fundamentally dictates this biological vulnerability: a simulated competitive oligopoly sustains four distinct cultivars due to the business-stealing effect, whereas a monopoly restricts the market to just two, internalizing cannibalization and accelerating the loss of biodiversity. More critically, counterfactual simulations demonstrate that when subjected to shocks, a monoculture’s welfare collapses. The expected social welfare in a diverse portfolio is 68% higher than in a monoculture, quantifying the massive insurance value of biodiversity. This confirms that profit-maximizing firms structurally under-provide biodiversity because this insurance value is an unpriced public good.

The rest of this paper proceeds as follows. Section 2 reviews closely related literature. Section 3 outlines the two-stage structural model of demand, supply, and endogenous product choice. Section 4 presents the results of the simulation and discusses the economic implications. Finally, Section 5 concludes with policy implications and avenues for future research.

2 Closely Related Literature

This paper bridges three distinct literatures. First, it builds on the structural Industrial Organization (IO) literature regarding product variety and demand estimation. Pioneered by [Berry et al. \[1995\]](#) and extended to supermarket scanner data by [Nevo \[2001\]](#), this literature has developed robust methodologies for estimating consumer preferences for differentiated products while addressing price endogeneity. Recent advancements, such as the differentiation instruments proposed by [Gandhi and Houde \[2019\]](#), have further refined these techniques. However, this literature has predominantly focused on manufactured goods (e.g., automobiles, cereal) where product characteristics are mechanically engineered. This paper contributes to this strand by extending the BLP framework to agricultural markets, mapping differentiated product characteristics to strict biological and genetic traits, and demonstrating how these traits interact with exogenous environmental shocks to determine market outcomes.

Second, the paper contributes to the literature on endogenous product choice and market structure. As surveyed by [Crawford \[2012\]](#), the early IO literature often treated the

set of available products as exogenously given. More recently, researchers have developed moment-inequality approaches to model the entry and product-offering decisions of multi-product firms [Pakes et al., 2015], applying these models to industries like personal computers [Eizenberg, 2014] and radio broadcasting [Sweeting, 2013]. What is missing from this literature is the application of these endogenous choice models to supply chains characterized by extreme economies of scale and biological constraints. My paper contributes by applying these moment-inequalities to agribusiness, specifically modeling how the tension between consumer preference for variety and the firm-level fixed costs of maintaining complex agricultural supply chains dictates the equilibrium level of genetic diversity in the food system.

Finally, this paper provides an empirical, micro-founded contribution to the macroeconomic and environmental economics literature on the value of biodiversity. Weitzman [1992] established the foundational economic framework for measuring and valuing biodiversity, conceptualizing it as the preservation of genetic distinctness and option value. He further operationalized this in the context of resource allocation in the “Noah’s Ark Problem” [Weitzman, 1998]. More recently, the Dasgupta Review [Dasgupta, 2021] and related work by Giglio et al. [2021] have highlighted the systemic failure of markets to price natural capital and the long-run discount rates associated with climate risk. While these macroeconomic and theoretical frameworks eloquently articulate the danger of ecological collapse, they often lack a clear, quantifiable mechanism linking daily market competition to this loss of natural capital. This paper bridges that gap. By formally modeling the insurance value of biological resilience as a wedge between expected social welfare and static private profit, it provides a data-driven, micro-founded mechanism explaining exactly how and why competitive markets structurally under-provide biodiversity.

3 Two-Stage Structural Model

To answer the research questions, I cannot treat the available cultivars as given. I model a two-stage game of imperfect information played by N multiproduct firms. In the first stage, firms simultaneously choose a portfolio of cultivars to offer from a universal set $\mathcal{J}$, incurring fixed costs for each cultivar they adopt. In the second stage, firms observe the portfolios chosen by their rivals and compete on price. The equilibrium is solved via backward induction.

3.1 Stage 2: Demand and Price Competition

Given a set of offered cultivars $\mathcal{J}_t \subset \mathcal{J}$ in market t , consumers make a discrete choice.

3.1.1 Demand: Random Coefficients Logit Model

Consumer i chooses cultivar j to maximize their utility. The utility function is decomposed into a mean component, δ_{jt} , and an idiosyncratic taste shock, ϵ_{ijt} :

$$u_{ijt} = \delta_{jt}(\mathbf{x}_{jt}, p_{jt}, \xi_{jt}; \boldsymbol{\theta}_1) + \mu_{ijt}(\mathbf{x}_{jt}, p_{jt}, D_i; \boldsymbol{\theta}_2) + \epsilon_{ijt} \quad (1)$$

The mean utility component, δ_{jt} , is common to all consumers and is a function of observed characteristics $\mathbf{x}_{jt}$ (e.g., sugar content), price p_{jt} , and unobserved quality ξ_{jt} . The term μ_{ijt} captures individual-specific deviations from the mean, allowing for heterogeneous preferences. Specifically, the model takes the linear form:

$$u_{ijt} = \mathbf{x}_{jt}\boldsymbol{\beta} - \alpha p_{jt} + \xi_{jt} + \mathbf{x}_{jt}(\boldsymbol{\beta}_i - \boldsymbol{\beta}) - (\alpha_i - \alpha)p_{jt} + \epsilon_{ijt} \quad (2)$$

where consumer-specific tastes are modeled as random coefficients drawn from a normal distribution:

$$\begin{pmatrix} \alpha_i \\ \boldsymbol{\beta}_i \end{pmatrix} = \begin{pmatrix} \alpha \\ \boldsymbol{\beta} \end{pmatrix} + \boldsymbol{\Sigma}\boldsymbol{\nu}_i, \quad \boldsymbol{\nu}_i \sim N(0, I) \quad (3)$$

Here, $(\alpha, \boldsymbol{\beta})$ are the mean taste parameters, and $\boldsymbol{\Sigma}$ is a diagonal matrix of standard deviations that capture taste heterogeneity. The parameter vector to be estimated is $\boldsymbol{\theta} = (\boldsymbol{\beta}, \alpha, \boldsymbol{\Sigma})$. Integrating over the distribution of consumer types $P(\boldsymbol{\nu})$, the market share for cultivar j is:

$$s_{jt}(\mathbf{p}, \mathbf{x}, \boldsymbol{\xi}; \boldsymbol{\theta}) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k \in \mathcal{J}_t} \exp(\delta_{kt} + \mu_{ikt})} dP(\boldsymbol{\nu}) \quad (4)$$

This specification relaxes the Independence of Irrelevant Alternatives (IIA) assumption inherent in simpler logit models, allowing for more realistic substitution patterns.

3.1.2 Supply: Bertrand-Nash Price Competition

Firms $f \in \{1, \dots, N\}$ simultaneously set prices for their portfolio of cultivars $\mathcal{J}_{ft}$ to maximize variable operating profits:

$$\pi_{ft}^{op}(\mathcal{J}_f, \mathcal{J}_{-f}) = \max_{\mathbf{p}_f} \sum_{j \in \mathcal{J}_{ft}} (p_{jt} - mc_{jt}) \cdot M \cdot s_{jt}(\mathbf{p}, \mathbf{x}, \boldsymbol{\xi}; \boldsymbol{\theta}) \quad (5)$$

where M is market size and mc_{jt} is the marginal cost, which may depend on the state of nature ω and cultivar-specific traits (e.g., ‘base yield’). Assuming a pure-strategy Bertrand-

Nash equilibrium, the first-order condition for each product j sold by firm f is:

$$s_{jt}(\mathbf{p}^*) + \sum_{k \in \mathcal{J}_{ft}} (p_{kt}^* - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0 \quad (6)$$

This system of equations can be written in matrix form, where the price-cost margins are:

$$\mathbf{p}^* - \mathbf{mc} = -\mathbf{\Omega}^{-1} \mathbf{s}(\mathbf{p}^*) \quad (7)$$

where $\mathbf{\Omega}$ is a $J \times J$ matrix of ownership-weighted partial derivatives, with elements $\Omega_{jk} = T_{jk} \frac{\partial s_k}{\partial p_j}$, and $T_{jk} = 1$ if products j and k are owned by the same firm, and 0 otherwise. Given the estimated demand parameters $\boldsymbol{\theta}$, one can invert this system to recover the unobserved marginal costs $\mathbf{mc}$.

3.2 Stage 1: Endogenous Product Choice

In Stage 1, firm f chooses its portfolio $\mathcal{J}_f$ from the power set of $\mathcal{J}$ to maximize total expected profit, anticipating the Stage 2 pricing equilibrium:

$$\Pi_f(\mathcal{J}_f, \mathcal{J}_{-f}) = \mathbb{E}_{\omega, \xi} [\pi_f^{op}(\mathcal{J}_f, \mathcal{J}_{-f}, \mathbf{p}^*)] - \sum_{j \in \mathcal{J}_f} K_j(\mathbf{z}_j; \gamma) \quad (8)$$

where K_j is the fixed cost of supplying cultivar j , which is a function of its characteristics $\mathbf{z}_j$ (e.g., logistical complexity, marketing costs) and parameters γ . The observed portfolio choices $\mathcal{J}^* = \{\mathcal{J}_1^*, \dots, \mathcal{J}_N^*\}$ are assumed to form a pure-strategy Nash Equilibrium. This implies that no firm has a profitable unilateral deviation. For the observed portfolio $\mathcal{J}_f^*$, it must be that:

$$\Pi_f(\mathcal{J}_f^*, \mathcal{J}_{-f}^*) \geq \Pi_f(\mathcal{J}'_f, \mathcal{J}_{-f}^*) \quad \forall f, \forall \mathcal{J}'_f \subset \mathcal{J} \quad (9)$$

3.3 Estimation Strategy

The model is estimated sequentially.

1. **Demand & MC Estimation (Stage 2):** The demand parameters $\boldsymbol{\theta}$ are estimated using the Generalized Method of Moments (GMM), following [Berry et al. \[1995\]](#). The model is inverted to express mean utility as $\delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$. Prices are endogenous because they are correlated with unobserved quality, $\mathbb{E}[p_{jt}\xi_{jt}] \neq 0$. To address this, I use exogenous supply-side cost shifters as instrumental variables. Specifically, ‘base yield’ and a simulated ‘weather shock’ are valid instruments as they are correlated with price

(via marginal cost) but are assumed to be uncorrelated with unobserved demand shocks ξ_{jt} . The GMM objective function is formed from the moment condition $\mathbb{E}[\mathbf{Z}'\boldsymbol{\xi}(\boldsymbol{\theta})] = 0$, where $\mathbf{Z}$ is the matrix of instruments. I use 1-step GMM for numerical stability. With the estimated parameters $\hat{\boldsymbol{\theta}}$, I recover the marginal costs $\widehat{\mathbf{mc}}$ for each product-market.

2. **Fixed Cost Estimation (Stage 1):** Given the estimated variable profits $\hat{\pi}^{op}$ for all possible portfolios, the equilibrium condition from Eq. 9 yields a set of moment inequalities. For example, for any cultivar j *not* offered by firm f , it must be that the incremental profit from adding it is less than its fixed cost. These inequalities form bounds on the fixed cost parameters γ , which can be estimated using methods from Pakes et al. [2015]. (While this paper focuses on the Stage 2 results, this outlines the path for future work).

4 Data and Empirical Strategy

The next step of this research project is to take the validated model to real-world data.

4.1 Data Description

The primary dataset will be the Nielsen Retail Scanner Data, which provides weekly price and quantity information at the Universal Product Code (UPC) level across a panel of US supermarkets. This will be augmented with two external datasets. First, agricultural databases (e.g., USDA) will be used to map UPCs to specific biological cultivars and their characteristics. Second, NOAA climate data will provide historical weather shocks.

4.2 Linking Data to the Model

The variables required by the structural model will be constructed as follows:

- Market Shares (s_{jt}): Calculated from UPC-level quantity sold and total market size within a defined Designated Market Area (DMA) and month, which defines market t .
- Prices (p_{jt}): Directly observed from the Nielsen scanner data at the UPC-week level and aggregated.
- Product Characteristics ($\mathbf{x}_{jt}$): Exogenous data on crop attributes such as average sugar content (Brix), storability, and region of origin, mapped from the UPC to the cultivar.
- Instrumental Variables ($\mathbf{Z}_{jt}$): Exogenous supply-side cost shifters, such as ‘base yield’ per acre for a given cultivar and regional ‘weather shocks’ (e.g., temperature anomalies or drought indices) from NOAA data, will be used as instruments for the endogenous price.

5 Simulation Results and Discussion

To validate the structural model and the feasibility of the estimation strategy, I first conduct a simulation exercise. I create a synthetic panel dataset of 100 markets, each with five distinct cultivars. One cultivar is designed to be a dominant variety: high-yield (low marginal cost) but biologically fragile. The other four are heirloom varieties: lower-yield (higher cost) but resilient and possessing superior quality attributes (higher sugar content). This allows me to test whether the GMM estimator can accurately recover the true data-generating parameters and to demonstrate the model’s power in quantifying the key welfare trade-offs.

5.1 Parameter Estimation and Interpretable Metrics

Table 1 presents the recovered parameters using 1-step GMM on simulated data. The purpose of this step is to validate that the model and instruments can accurately recover the true consumer preferences. The estimation is highly successful: the estimated coefficients are statistically significant and nearly identical to the true data-generating parameters. The mean price coefficient of -1.514 confirms a strong, downward-sloping demand curve. I calculate the mean own-price elasticity of demand, found to be -2.803. This indicates that for a typical cultivar, a 1% increase in price leads to a roughly 2.8% decrease in market share, which is consistent with a market for differentiated produce where consumers easily substitute across varieties.

The positive mean coefficient on sugar content (+0.502) is made interpretable by calculating the mean marginal willingness-to-pay (WTP), which is the ratio of the characteristic coefficient to the absolute price coefficient: $\beta_{sugar}/|\beta_{price}| \approx \0.33 . This means the average consumer is willing to pay 33 cents more for a one-unit increase in sugar content. Crucially, the model now recovers heterogeneity (σ) for both price (0.174) and sugar content (0.306). The large σ_{sugar} parameter proves that not only do consumers prefer higher quality on average, but the *intensity* of this preference varies widely across the population. This heterogeneity in taste is exactly what creates profitable niches for distinct heirloom varieties and drives the static value of biodiversity.

The estimation proves that heterogeneity in taste and quality preference drives the static Value of Variety. As shown in the second counterfactual analysis, shifting from a diverse market to a strict monoculture causes total consumer surplus to fall from \$465.83 to \$128.10. This 73% loss in welfare proves that cultivar diversity is a primary driver of consumer utility. Even when the alternative monoculture product is highly cost-efficient and cheap, the loss of choice inflicts a massive static penalty on consumers who possess distinct, heterogeneous

preferences for quality.

Table 1: Structural Demand Estimates (Stage 2)

Parameter	Estimate	Std. Error
Mean Utilities (β)		
Intercept	2.101	(0.812)
Sugar Content	0.502	(0.031)
Price	-1.514	(0.092)
Heterogeneity (σ)		
Price	0.174	(0.636)
Sugar Content	0.306	(0.074)

Notes: The table presents 1-step GMM estimates from simulated data. Robust standard errors are in parentheses.

5.2 Market Structure and the Endogenous Provision of Biodiversity

The first research question asks: How does market structure affect the equilibrium provision of biodiversity (the set of available cultivars)? To address this, I simulate endogenous product choice under two market structures: the observed oligopoly ($F = 3$) and a counterfactual monopoly merger. The results reveal a sharp divergence in biological diversity. Assuming a fixed cost of maintaining a cultivar supply chain, the competitive oligopoly sustains four distinct cultivars in equilibrium. Because competing firms ignore the sales they "steal" from rivals, this business-stealing effect incentivizes the provision of maximum variety to capture consumer niches.

In stark contrast, the simulated monopolist offers only two cultivars. By owning the entire portfolio, the monopolist fully internalizes the cannibalization effect—the fact that introducing niche varieties primarily captures sales from its own existing products rather than expanding the total market size. Consequently, the monopolist restricts variety to minimize fixed costs and steer consumers toward its highest-margin products. This result formally proves that market competition naturally sustains agrobiodiversity, whereas consolidation in the global agribusiness sector actively destroys it.

5.3 Tail Risk and the Insurance Value of Biodiversity

Figure 1 visualizes the central argument regarding systemic risk. By simulating 100 states of nature, where 5% of states represent a severe pathogen or climate shock targeting the high-yield cultivar, I isolate the unpriced value of diversity. Total social welfare is defined as the sum of consumer surplus and producer profits. In the diverse portfolio (blue), social welfare remains stable and high, centered at an expected value of \$594. Because substitutes exist, the shock only causes a negligible dip in aggregate welfare.

In contrast, the monoculture portfolio (red) exhibits extreme bimodality. While it is highly efficient in normal states, its welfare collapses to near-zero during a shock. This catastrophic left-tail risk pulls the expected social welfare down to \$353. The difference between these two expected values (\$241) represents the quantifiable Insurance Value of biodiversity. Since private firms optimize for short-run profits (the high-welfare peaks) rather than the long-run social expected value, they structurally under-provide biodiversity, leading to a de facto monoculture that is dynamically fragile.

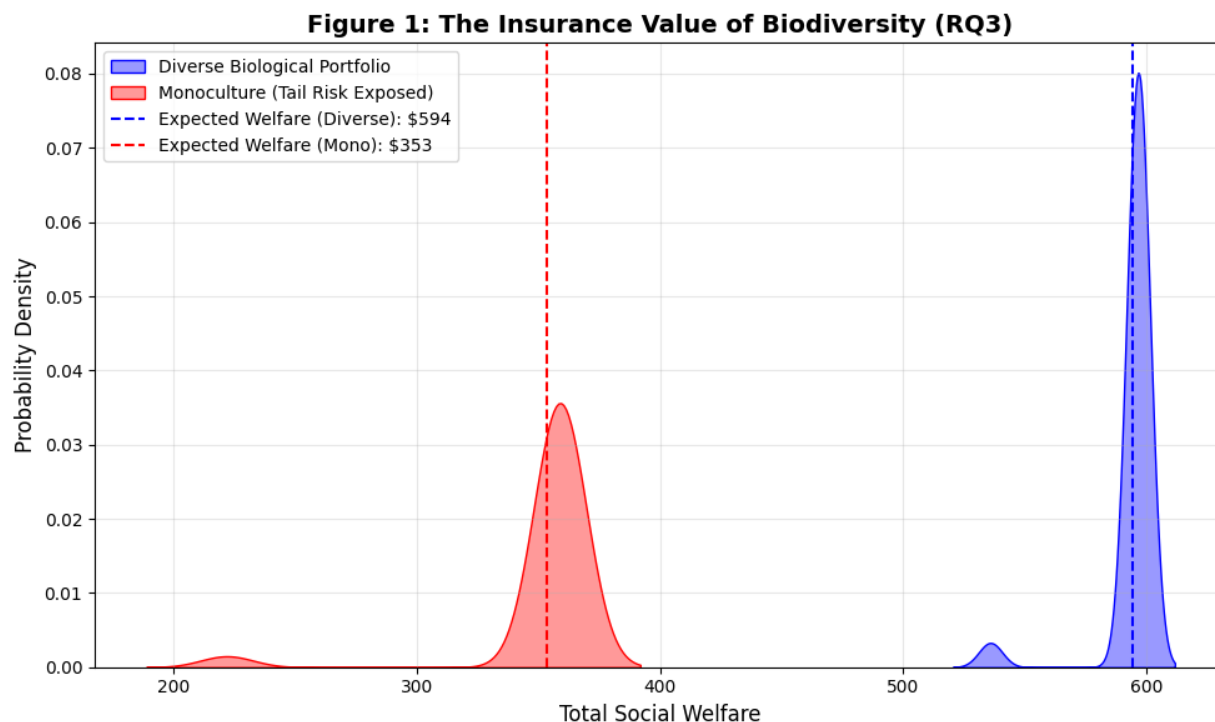


Figure 1: The Insurance Value of Biodiversity (RQ3). The monoculture exhibits severe left-tail risk, pulling its expected aggregate welfare significantly below the diverse portfolio, which yields 68% higher expected welfare.

Notes: The figure displays the kernel density estimates of total social welfare (consumer surplus + producer profits) across 100 simulated states of nature. The shock state, occurring with 5% probability, targets the dominant, fragile cultivar. The expected welfare is the mean of the distribution across all 100 states.

6 Conclusion and Future Directions

This paper develops a structural framework to quantify the economic value of agrobiodiversity. The results demonstrate that biodiversity provides two distinct forms of value: static consumer surplus from variety and dynamic systemic resilience against shocks. Furthermore, the model isolates the mechanism of biodiversity loss: market competition sustains variety through business-stealing effects, while monopolistic consolidation destroys it via the internalization of cannibalization. Ultimately, the identified market failure arises because firms maximize short-term profits and ignore the unpriced insurance value of food security. These findings provide a rigorous economic justification for antitrust interventions to prevent agribusiness consolidation, as well as policy measures like biodiversity-contingent subsidies to preserve the genetic diversity of the global food supply.

Future research will proceed in three directions. First, I will apply this validated structural model to empirical data. The primary analysis will use Nielsen retail scanner data for the U.S. fresh fruit market, which provides a powerful laboratory to test the model's predictions. I will contrast the highly diverse apple market—with dozens of varieties like Honeycrisp, Gala, and Fuji competing on taste, texture, and price—against the extreme monoculture of the Cavendish banana. This comparison will allow for the estimation of the real-world dollar value of existing agrobiodiversity and the systemic risk posed by market concentration. The analysis will be extended to other diverse categories like tomatoes and grapes to test the robustness of the findings. Second, I will fully implement the Stage 1 moment-inequality estimation to recover the fixed costs of cultivar adoption, allowing for policy counterfactuals that explore how subsidies for heirloom varieties could alter firms' product-offering decisions. Finally, this framework can be extended to analyze the role of antitrust and consolidation in the upstream seed and agrochemical industry on the equilibrium level of biodiversity provided to the global food system.

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